

5.2 Nervous control

Co-ordination may be chemical or electrical in nature. In this section, learners gain an understanding of the electrical component through the study of the structure and functions of the mammalian nervous system. The use of brain scanning techniques and traditional reflex tests to assess brain injury are investigated. An appreciation of the unique problems associated with damage to neurones is considered, as well as the effects of histological changes in the nervous system as a result of ageing.

The effects of drugs on the nervous system are also considered, including their use in the treatment of neurological conditions such as Parkinson's disease.

The eye is used to illustrate the role of the sensory receptor as a biological transducer. The use of optical scans and eye tests to assess visual function is also explored.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.

5.2.1 The nervous system and the identification and consequences of damage

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) (i) the organisation of the somatic and autonomic nervous system in humans	To include the use of diagrams, photomicrographs and electron micrographs to show the structure of the central and peripheral nervous system. <i>M0.1, M0.2, M1.1, M1.2, M1.8</i> PAG1 HSW4, HSW5, HSW6
(ii) practical observations of nervous tissue using a light microscope	
(b) the structure and function of the human brain	To include the use of diagrams, photomicrographs and electron micrographs of the locations of the cerebrum, cerebellum, medulla oblongata, hypothalamus and pituitary gland and the functions of each area. <i>M0.1, M0.2, M1.1, M1.2, M1.8</i>
(c) the structure of motor, sensory and relay neurones	To include the use of diagrams, photomicrographs and electron micrographs to show axons, dendrons and dendrites, cell surface membrane and cell body, myelin sheath and nodes of Ranvier. <i>M0.1, M0.2, M1.1, M1.2, M1.8</i>

(d) the establishment of the resting potential and the transmission of the action potential in neurones	To include the interpretation of graphs showing changes in membrane potential, the significance of the myelin sheath and the refractory period in the transmission of nerve impulses. <i>M3.1, M3.4</i>
(e) the structure and function of synapses in integrating responses in the nervous system	To include the role of neurotransmitters in excitatory and inhibitory post synaptic potential.
(f) (i) the nature of a reflex arc, the use of reflexes and the differences when compared to a reaction (ii) practical investigations into reflexes in humans (iii) practical investigations into factors affecting reaction times	Reflexes to include the use of the blink reflex, iris reflex and plantar reflex in assessing nervous system damage and levels of consciousness AND the use of the Student's <i>t</i> -test in investigations of the factors affecting reaction time. <i>M0.1, M0.3, M1.1, M1.2, M1.3, M1.6, M3.1, M3.2, M3.4, M3.5</i> PAG11 HSW3, HSW4, HSW5, HSW6
(g) the use of brain scans in assessing brain and spinal cord damage	To include the assessment of traumatic brain injury and strokes and scans to include CT, MRI, fMRI and PET scans and EEGs. HSW9, HSW10, HSW11, HSW12
(h) the consequences of brain and spinal cord damage	To include effects on memory, motor skills, speech and hormonal imbalance and the ethical consequences of establishing brain death should be discussed. HSW10
(i) the use of drugs to modify brain activity and function	To include the therapeutic use of drugs, such as dopamine for the treatment of Parkinson's disease and the effect of heroin, cannabis, methamphetamine and alcohol on synapse activity in the brain. HSW5, HSW6, HSW7, HSW9, HSW10, HSW11, HSW12
(j) psychological and physical drug dependency.	To include the biological basis of dependency and the consequences of dependency on individuals and society. HSW10, HSW12